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GitHub Link	**https://github.com/Nagellajeevansai/MLA0405-Deep-Learning.git**

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 2

Game Based Learning

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0405**

SLOT : D

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

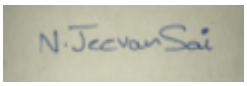
CO Weightage: 25% **Duration: 1 Hour**

Assessment Tool Weightage: 35% **Marks: 50**

Code of Conduct:

IN.Jeevan Sai(192525169)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student: **Name of the student:N.Jeevan Sai**

Criteria	Understanding of Concept **(2.5)**	Conceptual Clarity **(2.5)**	ProblemSolving Ability **(2)**	Solution Design & Application **(2)**	Teamwork & Game Participation ***(1)***	**Total (10)**
Weightage	***25 %***	***25%***	***20%***	***20 %***	***10 %***	**100%**
Round 1						
Round 2						

Total						

Signature of the Faculty: ***Total Marks :***

QUESTIONS:01 **Total Marks: 50**

Question:

Design and participate in a two-round game-based learning activity to explore the fundamental concepts of Convolutional Neural Networks (CNNs), including architecture, layers, filters, parameter sharing, regularization, AlexNet, ResNet, and real-world applications. Progress through five levels in each round, applying your knowledge to solve challenges involving CNN design, feature extraction, architecture comparison, and practical applications.

The Game Progression

Round 1:

Basic CNN Concepts → Filters → Parameter Sharing → Architecture → AlexNet vs ResNet **Round 2:**

Convolution → Feature Maps → Regularization → Architecture Comparison → RealWorld Application

ROUND 1: CNN Architecture Builder (Learn by constructing and understanding CNN architectures layer by layer)

Level	Challenge	Description
Level 1	**CNN Layer Matcher**	Match CNN layers such as Convolution, Pooling, Flatten, and Fully Connected layers with their respective functions and roles in the network.
Level 2	**Filter Finder**	Identify suitable filters and kernels for feature extraction and understand how filter size, stride, and padding affect the output feature map.
Level 3	**Parameter Sharing Puzzle**	Determine how the same filter weights are reused across different image locations and calculate the number of parameters in a convolutional layer.
Level 4	**CNN Architecture Debugger**	Identify and correct problems in a CNN architecture, such as incorrect layer order, incompatible dimensions, and inappropriate layer selection.
Level 5	**AlexNet vs ResNet Challenge**	Compare AlexNet and ResNet architectures and identify how their architectural approaches differ, particularly the use of residual connections in ResNet.

ROUND 2: CNN Feature Detective (Learn by exploring how CNNs extract features, avoid overfitting, and solve real-world problems)

Level	Challenge	Description
Level 1	**Edge Detection Explorer**	Apply simple (3\times3) filters to an image and determine how convolution operations can detect edges and other basic features.
Level 2	**Feature Map Detective**	Analyze feature maps produced by different CNN layers and identify the progression from simple features such as edges to complex patterns and objects.
Level	**Challenge**	**Description**
Level 3	**Regularization Rescuer**	Identify overfitting in a CNN model and select suitable regularization techniques to improve generalization and model performance.
Level 4	**CNN Architecture Race**	Compare ResNet and AlexNet for different image-classification applications and determine which architecture is more suitable for a given problem.
Level 5	**CNN Application Master**	Design a CNN-based solution for a real-world application such as medical image analysis, face recognition, object detection, or autonomous driving, and justify the selected architecture and layers.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Conceptual Clarity	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Problem-Solving Ability	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design & Application	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Teamwork & Game Participation	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CNN GAME-BASED LEARNING ACTIVITY

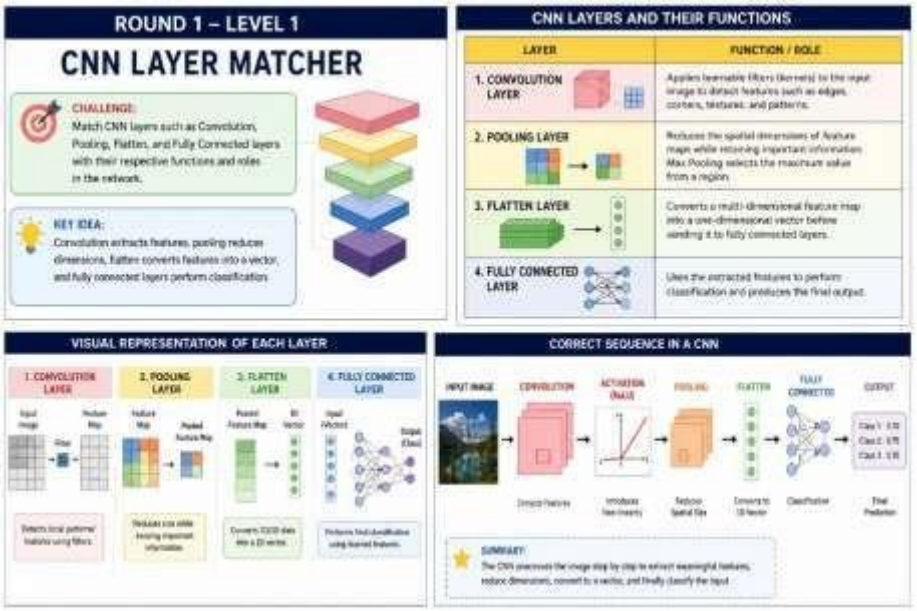
ROUND 1: CNN Architecture Builder

Level 1 – CNN Layer Matcher

CNN Layer	Function / Role
Convolution Layer	Applies learnable filters/kernels to the input image to detect features such as edges, corners, textures, and patterns.
Pooling Layer	Reduces the spatial dimensions of feature maps while retaining important information. Max Pooling selects the maximum value from a region.
Flatten Layer	Converts a multi-dimensional feature map into a one-dimensional vector before sending it to fully connected layers.
Fully Connected Layer	Uses the extracted features to perform classification and produces the final output.

Correct sequence:

Input Image → Convolution → Activation (ReLU) → Pooling → Flatten → Fully Connected → Output



Key idea: Convolution extracts features, pooling reduces dimensions, flatten converts features into a vector, and fully connected layers perform classification.

Level 2 – Filter Finder

A CNN uses small kernels/filters to detect different features in an image.

For example, a **3 × 3 edge-detection filter** can be:

This filter detects **horizontal edges**.

Another filter:

can detect **vertical edges**.

Effect of filter parameters

- **Filter size:** Determines the region examined at each convolution step.
- **Stride:** Determines how far the filter moves at each step.
- **Padding:** Adds pixels around the image to control the output size.
- **Number of filters:** Determines the number of feature maps generated.

The output size is: where:

- = input size
- = filter size
- = padding
- = stride

Example

For a image, filter, stride , and padding :

Therefore, the output feature map is:

Level 3 – Parameter Sharing Puzzle

CNNs use **parameter sharing**, meaning that the same filter weights are reused at different positions of an image.

For example, consider a convolution layer with:

- Filter size =
- Input channels = 3
- Number of filters = 32
- One bias per filter

Number of parameters:

Therefore, the convolution layer contains **896 trainable parameters**.

Why parameter sharing is important

Parameter sharing:

1. Reduces the number of parameters.
2. Reduces computational requirements.
3. Helps detect the same feature at different image locations.
4. Improves CNN efficiency.

Example: A filter that detects an edge in the top-left corner can also detect the same type of edge in the center or bottom-right region.

Level 4 – CNN Architecture Debugger

Consider the following incorrect architecture:

Input → Flatten → Convolution → Pooling → Fully Connected

Problems

1. **Flatten is placed too early.**
    - Flatten converts the feature map into a 1D vector.
    - A convolution layer expects spatial information.
2. **Convolution should come before Flatten.**
    - CNNs need spatial information to extract image features.
3. **Pooling should normally follow convolution/activation. Correct architecture**

Input Image

↓

Convolution

↓

ReLU Activation

↓

Pooling

↓

Convolution

↓

ReLU

↓

Pooling

↓

Flatten

↓

Fully Connected

↓

Output

Debugging conclusion

The major problem was the incorrect placement of the Flatten layer. The CNN should first extract spatial features using convolution and pooling and only then flatten the feature maps for classification.

Level 5 – AlexNet vs ResNet Challenge

Feature	**AlexNet**	**ResNet**
Introduced	2012	2015
Main idea	Deep CNN with convolution and pooling	Deep CNN using residual learning
Architecture	Conventional sequential architecture	Uses residual/skip connections
Activation	ReLU	ReLU
Important contribution	Demonstrated strong GPU-based deep learning performance	Solved degradation problems in very deep networks
Depth	8 learned layers	Can have 18, 34, 50, 101, 152 layers, etc.
Skip connections	No	Yes
Training very deep networks	Difficult	Easier
Main advantage	Historically important and comparatively simple	Effective for very deep networks

Residual connection

ResNet introduces a shortcut connection:

Instead of learning the complete transformation directly, the network learns a **residual function** .

Conclusion

AlexNet was a major breakthrough in deep CNNs, while **ResNet** introduced residual connections that made training much deeper networks practical.

ROUND 2: CNN Feature Detective

Level 1 – Edge Detection Explorer

Consider the following input:

Use the horizontal edge filter:

Element-wise multiplication and summation:

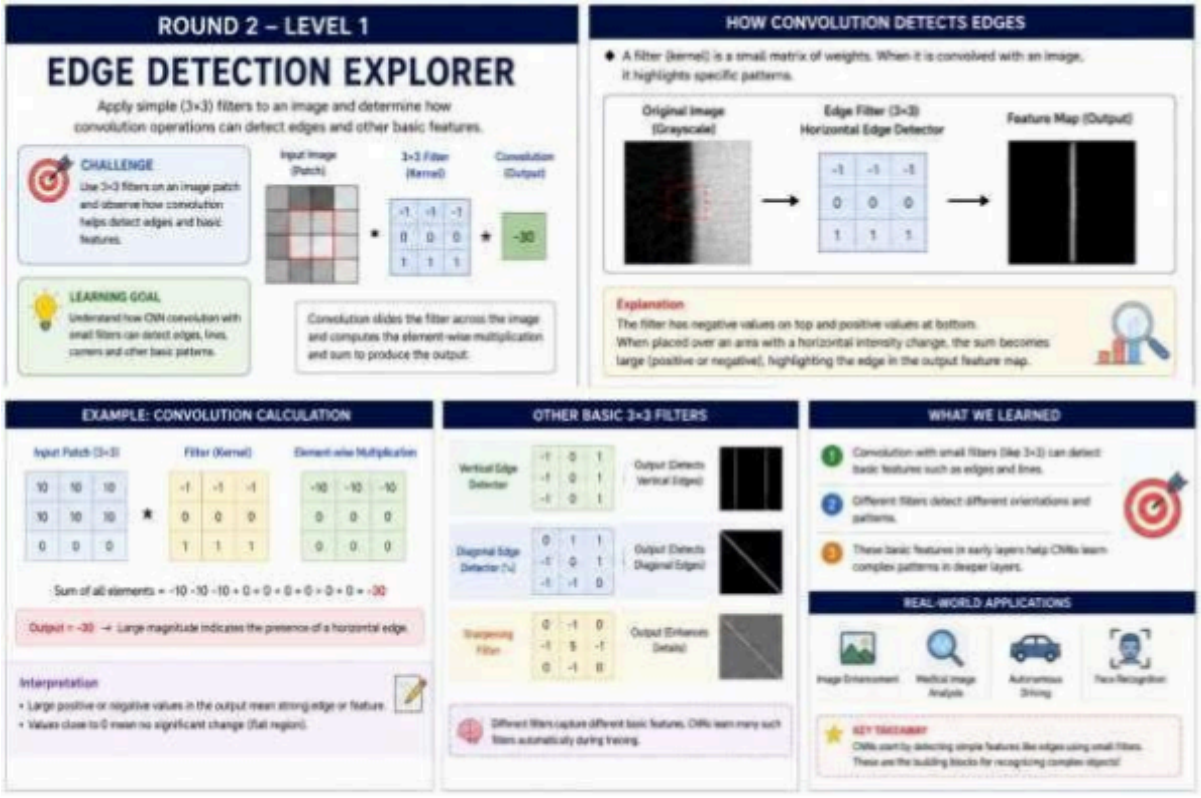
The large magnitude indicates a strong **horizontal edge**.

Interpretation

CNN filters detect basic visual patterns such as:

- Horizontal edges
- Vertical edges
- Diagonal edges
- Corners
- Textures

These basic features are combined by deeper layers to identify complex objects.



Level 2 – Feature Map Detective

CNN layers progressively learn more complex features.

Early layers Detect:

- Edges
- Lines
- Corners
- Simple textures **Middle layers** Detect:
- Shapes
- Curves
- Object parts

- Patterns

Deep layers Detect:

- Faces
- Animals
- Vehicles
- Objects
- Complex structures

Example

For a face-recognition CNN:

Layer 1 → Edges

↓

Layer 2 → Eyes, nose, mouth shapes

↓

Layer 3 → Face components

↓

Deep layers → Complete face representation

Therefore, CNN feature extraction progresses from **simple features to complex semantic features**.

Level 3 – Regularization Rescuer

Problem

Suppose a CNN produces:

- Training accuracy = **98%**
- Validation accuracy = **75%**

This indicates **overfitting**.

The model performs extremely well on training data but poorly on unseen data.

Suitable solutions

1. Dropout

Randomly disables some neurons during training.

Example:

Dropout = 0.5

Approximately 50% of selected neurons are randomly ignored during a training step. **2.**

Data Augmentation

Creates additional variations of training images through:

- Rotation
- Flipping
- Cropping
- Scaling
- Brightness changes

3. L2 Regularization

Adds a penalty for large weights to the loss function:

4. Early Stopping

Training is stopped when validation performance stops improving.

Recommended solution

For an image classification problem, a combination of: **Data**

Augmentation + Dropout + Early Stopping

can effectively reduce overfitting.

Level 4 – CNN Architecture Race

Suppose two applications are given:

Application A: Small image classification dataset

Recommended architecture: AlexNet-style CNN

Reason:

- Relatively simpler architecture.
- Suitable for learning basic image features.
- Easier to train compared with very deep networks.

Application B: Large-scale complex image classification

Recommended architecture: ResNet Reason:

- Supports very deep networks.
- Residual connections improve gradient flow.
- Can learn complex hierarchical features.
- Generally more suitable for challenging image classification tasks.

Comparison

Requirement	**AlexNet**	**ResNet**
Simple architecture	✓	
Very deep network		✓

Residual/skip connections		✓
Complex feature learning	Good	Excellent
Deep network training	More difficult	Easier
Modern image classification	Less preferred	More preferred

Winner

For most modern complex image-classification applications:

is the better choice.

Level 5 – CNN Application Master

Application: Medical X-Ray Image Classification

Problem

Develop a CNN that classifies chest X-ray images into:

- **Normal**
- **Disease affected**

Proposed architecture

Chest X-ray

↓

Image Preprocessing

↓

Convolution Layer

↓

ReLU

↓

Max Pooling

↓

Convolution Layer

↓

ReLU

↓

Max Pooling

↓

Residual/Deep CNN Layers

↓

Global Average Pooling

↓

Fully Connected Layer

↓

Sigmoid Output

↓ Normal /

Disease

Why ResNet?

ResNet is suitable because:

1. It can learn complex visual patterns.
2. Residual connections allow deeper networks to be trained effectively.
3. It can extract hierarchical features from X-ray images.
4. Transfer learning can be used when the medical dataset is relatively small.

Output

For binary classification:

If:

the image can be classified as **Disease**.

Otherwise:

the image is classified as **Normal**.

Real-world applications of CNNs CNNs

can be applied to:

- Medical image analysis
- Face recognition
- Autonomous driving
- Traffic sign recognition · Object detection
- Satellite image analysis
- Industrial defect detection

Final Game Learning Summary

Round	**Level**	**Main Concept**	**Key Learning**
Round1	1	CNN Layers	Understand the role of each layer
	2	Filters	Learn feature extraction using kernels
	3	Parameter Sharing	Calculate and understand shared weights

	4	Architecture	Debug incorrect CNN designs
	5	AlexNet vs ResNet	Understand conventional vs residual architectures
Round2	1	Convolution	Detect edges using filters
	2	Feature Maps	Understand simple-to-complex feature extraction
	3	Regularization	Prevent overfitting
	4	Architecture Comparison	Select suitable CNN architectures
	5	Real-World Application	Design a CNN solution for practical problems

Overall conclusion

The two-round game demonstrates the complete CNN learning process, beginning with basic layers and filters and progressing toward architecture design, parameter calculation, feature extraction, regularization, architecture comparison, and real-world applications. Through the activities, learners understand why CNNs are highly effective for image-based tasks and why architectures such as ResNet are important for deep learning.

Presentation Pic